

The residual, bounded at every instant

drop-in for Sec. VIII; the RMS form is (VIII.3)

0.1 What is to be proved, and what is not true

Write the recorded rate over the post-onset window as the nominal response, the deviation the model does not carry, and sensor noise,

$$y(\tau) = \omega_{\text{nom}}(\tau) + e_\omega(\tau) + n(\tau), \quad (\text{P.1})$$

and let $\hat{\omega}$ be the fitted curve. The natural statement to want about the residual $r = y - \hat{\omega}$ is

$$|r(\tau)| \stackrel{?}{\leq} |e_\omega(\tau)| + k\sigma_n. \quad (\text{P.2})$$

It is false, and the counterexample is at the onset. The envelope of (93) vanishes there, $E(\tau) \propto \sinh C_2\tau \rightarrow 0$, and so does e_ω ; the residual does not. The fitted curve is selected by the whole window, so a deviation living at large τ displaces the fit at $\tau = 0$ as well, and (P.2) would require that displacement to vanish where e_ω does. Measured on the campaign, the ratio $(E + D)/E$ of Sec. 0.2 below is 2.07 to 2.15 at the end of the window and 27 to 95 at the first sample: the multiplicative form has no uniform constant.

What is missing from (P.2) is the displacement of the fit itself. Naming it costs one term and makes the statement provable.

0.2 The residual is exactly a projection

The identification runs in two stages and they must not be confused here. Stage one estimates (C_2, K) for a configuration by a *free* per-run nonlinear least squares followed by a per-configuration median, refined on a ramp-invariance criterion; that fit is nonlinear and this section says nothing about it. Stage two — the per-run onset fit whose residual Sec. VIII checks — takes those two as *given constants*. Everything below is a statement about stage two, conditional on the calibration it is handed, and Sec. 0.5 pays for that conditioning.

With C_2 a constant the model is *linear* in the amplitude and the pre-onset level, so on the sample grid $\tau_1, \dots, \tau_N$ of the window the admissible curves form a genuine two-dimensional subspace

$$V = \text{span}\{u, \mathbf{1}\} = \text{range } A, \quad A = \begin{bmatrix} u & \mathbf{1} \end{bmatrix} \in \mathbb{R}^{N \times 2}, \quad (\text{P.3})$$

with $u_i = \cosh C_2\tau_i - 1$. This is the one structural fact the whole argument needs, and it is worth stating explicitly because the estimator is elsewhere described as a nonlinear fit: it is nonlinear in C_2 , which stage one estimates and stage two does not touch, and in the onset, which is swept rather than minimised over analytically. In the parameters that remain it is linear.

Let $P = A(A^\top A)^{-1}A^\top$ be the orthogonal projector onto V , so that the least-squares fit is $\hat{\omega} = Py$.

The constant $\mathbf{1}$ belongs in the span because the fitted curve carries an additive pre-onset level: (98) minimises over both segments with the same offset C , and the post-onset model is $\hat{\omega}(t; t_c) + C$. That level is *not* gyro bias — the rate comes from the navigation filter, whose delta-angle bias states remove sensor bias — but the residual motion the vehicle still has when the excitation begins, measured here at 0.004 to 0.047°/s. What V may contain is dictated by what the estimator ranges over and not chosen for convenience: writing $V = \text{span}\{u\}$ while the code also fits a level would make $\hat{\omega} \neq Py$ and Lemma 1 false.

The second dimension is close to free. Dropping $\mathbf{1}$ would take $\text{tr } P$ from 2 to 1 and the constant of (P.7) from 2.07–2.15 down to 2.04–2.12, a gain of 1.2%; keeping it annihilates the constant channel entirely, and a constant in a direction u cannot mimic at all — u vanishes at the onset, so a level b would survive into the residual with a factor of 1.000.

Lemma 1 (the nominal cancels). $\omega_{\text{nom}} \in V$, being C_1^*u for the true amplitude, so $P\omega_{\text{nom}} = \omega_{\text{nom}}$ and

$$r = (I - P)y = (I - P)(e_\omega + n). \quad (\text{P.4})$$

The nominal response has disappeared entirely: the residual is a function of the perturbations alone, exactly and with no approximation. This is the step at which the intuition behind (P.2) becomes an identity, and everything after it is bookkeeping on a two-dimensional projector.

Lemma 2 (the deviation channel). By (93) the deviation is confined to the band $|e_{\omega,j}| \leq E_j$. A linear functional over a box attains its extremum at a vertex, so for each i

$$|(Pe_{\omega})_i| \leq D_i \triangleq \sum_j |P_{ij}| E_j, \quad (\text{P.5})$$

with equality for $e_{\omega,j} = E_j \operatorname{sgn} P_{ij}$. The bound is sharp over the band, though conservative over the deviations that are actually reachable, those being Duhamel images of an admissible ρ and therefore smooth.

Lemma 3 (the noise channel). For $n \sim \mathcal{N}(0, \sigma_n^2 I)$ and P symmetric idempotent, $\operatorname{Var}[(Pn)_i] = \sigma_n^2 (P^2)_{ii} = \sigma_n^2 P_{ii}$, so $|(Pn)_i| \leq k \sigma_n \sqrt{P_{ii}}$ at confidence $2\Phi(k) - 1$.

Theorem. Combining (P.4) with the two lemmas,

$$|r(\tau_i)| \leq \underbrace{E_i}_{\text{model}} + \underbrace{D_i}_{\substack{\text{displacement} \\ \text{of the fit}}} + k \sigma_n (1 + \sqrt{P_{ii}}), \quad (\text{P.6})$$

and, writing $\Lambda_E = \max_i D_i / \max_j E_j$ and $\kappa = \max_i \sqrt{P_{ii}}$, in uniform form

$$\max_i |r_i| \leq (1 + \Lambda_E) \sup_{\tau} E(\tau) + (1 + \kappa) k \sigma_n. \quad (\text{P.7})$$

Two Gaussian events enter per sample, n_i and $(Pn)_i$, so a union bound over the window replaces k by $\Phi^{-1}(1 - \alpha/4N)$: for $N = 53$ samples that is $k = 3.50$ at 95% and 3.90 at 99%, against 1.96 and 2.58 for a single sample.

0.3 The constants, and why they do not grow

Everything in (P.6) is computable before a run is flown: P depends only on C_2 and the sample grid, E only on the geometry box.

P itself is the standard least-squares projector. Minimising $\|y - A\theta\|^2$ gives the normal equations $A^\top A \hat{\theta} = A^\top y$, whose content is $A^\top (y - A\hat{\theta}) = 0$: the residual is orthogonal to every column of A , which is to say to everything the model can produce. With A of full column rank, $\hat{\theta} = (A^\top A)^{-1} A^\top y$ and $\hat{\omega} = A\hat{\theta} = Py$. For the two columns here it collapses to a form with no matrix inverse left in it,

$$P_{ij} = \frac{1}{N} + \frac{\tilde{u}_i \tilde{u}_j}{\sum_k \tilde{u}_k^2}, \quad \tilde{u} = u - \bar{u}, \quad (\text{P.9})$$

which is the constant channel plus the u channel with the constant removed, and it reproduces the numerically formed P to 10^{-16} . Every property the proof uses is one line from it. $P^2 = P$ and $P^\top = P$ are immediate; $Pv = v$ for $v \in V$ is Lemma 1; $\operatorname{Var}[(Pn)_i] = \sigma_n^2 (PP^\top)_{ii} = \sigma_n^2 P_{ii}$ is Lemma 3; and

$$\operatorname{tr} P = \sum_i \frac{1}{N} + \frac{\sum_i \tilde{u}_i^2}{\sum_k \tilde{u}_k^2} = 1 + 1 = 2,$$

one unit for each basis direction, whatever N is. Equation (P.9) also locates the worst sample: P_{ii} is largest where $|\tilde{u}_i|$ is, which is the last sample of the window, $\tau/\tau_{\text{end}} = 1.000$ in every one of the 140 runs. There P_{ii} reaches 0.124 to 0.164, of which only $1/N = 0.013$ to 0.024 is the constant — the rest is the amplitude direction, whose cosh has grown by then. This is why κ is 0.35 to 0.40 rather than the $\sqrt{2/N} \approx 0.2$ a flat basis would give.

$\dot{M}$ N m/s	N	$\operatorname{tr} P$	Λ Lebesgue	$1 + \Lambda_E$ sharp	κ	$1 + \kappa$
0.10	74	2.000	2.104	2.066	0.352	1.352
0.20	63	2.000	2.046	2.089	0.362	1.362
0.30	57	2.000	2.012	2.101	0.370	1.370
0.45	54	2.000	1.983	2.113	0.376	1.376
0.65	48	2.000	1.962	2.130	0.385	1.385
0.90	44	2.000	1.937	2.140	0.395	1.395
1.20	41	2.000	1.924	2.149	0.404	1.404

The trace is 2 at every rate and every window length, being the dimension of the range, and that is the reason the constants do not grow with the record: a two-dimensional projector cannot move much, however many samples it is given. The Lebesgue constant $\Lambda = \max_i \sum_j |P_{ij}|$, which is what (P.5) reduces to if E is replaced by its supremum, sits between 1.92 and 2.10 across a twelvefold change in ramp rate. Carrying the actual shape of E instead gives $1 + \Lambda_E \simeq 2.1$, about a third smaller. Over all 140 runs

$$\max |r| \leq 2.18 \sup E + 1.48 k \sigma_n.$$

0.4 The bound tested, and the two things it exposes

Applied to every sample of every run at $k = 3.5$:

$\dot{M}$ N m/s	runs entirely inside σ_{pre}	worst σ_{in}	ratio	deployed worst	$\sigma_{\text{pre}} / \sigma_{\text{in}}$ °/s
0.10	19/20	20/20	0.433	0.552	0.087 / 0.831
0.20	12/20	20/20	0.445	0.839	0.044 / 0.800
0.30	13/20	20/20	0.575	0.830	0.101 / 0.823
0.45	12/20	20/20	0.685	1.350	0.127 / 0.707
0.65	7/20	20/20	0.754	1.045	0.122 / 0.856
0.90	4/20	20/20	0.763	1.210	0.111 / 0.749
1.20	7/20	20/20	0.714	1.524	0.173 / 0.773

The theorem holds on all 140 runs at every sample, the worst ratio being 0.763. Two conditions on how it is applied are not optional, and the table makes both visible.

The noise scale must be measured in the window. Taking σ_n from the pre-onset record leaves only 74 of 140 runs inside. That is not the theorem failing but its hypothesis: before the onset the vehicle is carried by every landing gear, after it pivots on one edge, and the in-window disturbance is 0.71 to 0.86°/s against a pre-onset floor of 0.04 to 0.17. The honest estimate is the in-window content above 5 Hz, which the fitted model provably cannot produce — $\hat{\omega}$ is smooth on the scale of the window — so whatever is there is disturbance by construction rather than by assumption.

The deployed fit is more constrained, and each constraint adds a term. V is the family the bound is about: amplitude and level free. The shipped estimator frees neither. It takes $C_1 = KM$ from the calibration stage, and it forms C from the *pre-onset* segment rather than minimising over it on the window, so once the onset is chosen its post-onset curve has no free parameter at all and $\hat{\omega} \neq Py$. Lemma 1 then removes only part of the nominal and two terms survive,

$$|r_i| \leq |\Delta C_1| u_i + |C - b| + E_i + D_i + k \sigma_n (1 + \sqrt{P_{ii}}), \quad (\text{P.6}')$$

the first being the calibration's amplitude error and the second the baseline channel already carried through (105)–(106). Their measured sizes are very different. The amplitude term is $|\Delta C_1|/C_1 = 11.3\%$ and it is what puts 26 runs outside the RMS form (VIII.3); without it the deployed residual reaches 1.52 of the bound. The baseline term is $|C - b| \simeq \sigma_{\text{pre}}/\sqrt{N_{\text{pre}}} = 0.005$ to $0.029^\circ/\text{s}$, which is 0.1 to 1.8% of $\sup E$ (8.6% at worst over the 140 runs). One constraint matters and the other does not, and the bound says which.

0.5 What the calibration is worth: Lemma 1 needs C_2 exact

Lemma 1 removes the nominal because $\omega_{\text{nom}} \in V$, and V is built from the calibrated C_2 . If that value is not the true one the nominal leaves the subspace and the cancellation is only partial: with $C_2 = C_2^*(1 + \varepsilon_2)$ the residual retains $(I - P)\omega_{\text{nom}}$, and (P.6) gains a fourth term

$$|r_i| \leq |(I - P)\omega_{\text{nom}}|_i + |\Delta C_1| u_i + E_i + D_i + k \sigma_n (1 + \sqrt{P_{ii}}). \quad (\text{P.8})$$

This is the price of conditioning on stage one, and it is the reason the two stages had to be separated at the start. It is small, and almost linear in ε_2 :

$\dot{M}$	$\ (I - P)\omega_{\text{nom}}\ _{\infty} / \sup E$			
N m/s	$\varepsilon_2 = 1\%$	2%	5%	10%
0.10	0.005	0.010	0.027	0.059
0.45	0.013	0.027	0.071	0.152
0.90	0.018	0.036	0.093	0.195
1.20	0.020	0.041	0.105	0.221

A 2% exponent error costs 1 to 4% of the envelope and a 5% error costs 3 to 11%, the fast ramp being the more exposed because its window is shorter and the two hyperbolic shapes are less separated there. Like the pinned amplitude it enters additively and is not covered by $\bar{\rho}$, which bounds the physics the model omits and says nothing about the constants the model is given.

Note that the empirical check above is unaffected by any of this: it compares a *measured* record, which contains whatever exponent the rig actually has, against the envelope, so the 140/140 already includes the misspecification. What (P.8) supplies is the proof that this had to be so, and a budget line for it.

If C_2 is freed, as in stage one. V is then a two-dimensional manifold rather than a subspace and P is no longer a projector, so none of the above applies verbatim. The statement survives with a curvature factor: if the manifold has reach ϱ and $\|e_{\omega} + n\| < \varrho$, the displacement in (P.5) is multiplied by $(1 - \|e_{\omega} + n\|/\varrho)^{-1}$. That is the form to use for the calibration fit itself. For the per-run fit that Sec. VIII checks, C_2 is a constant and the linear form applies exactly, which is worth saying plainly: the projection argument is exact for the stage the residual test is about, not an approximation to it.